PATIENT MONITORING · PHYSIONET OPEN ICU DATA

Catching Patients Early Without Drowning the Ward in Alarms

A SQL-first look at early-warning alert rules on 1.55 million hours of vital signs from 40,336 ICU stays: how many sepsis patients each rule catches, how early, how many alarms it costs, and what a heart-rate + breathing-rate sensor alone can do.

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DATA
PhysioNet / CinC Challenge 2019 (CC BY 4.0)

TOOLS
SQL (DuckDB) · Python · pandas · matplotlib

Analyst	Mohammad Faiz Ahmad, IIT Patna	**Status**	Final
Data source	PhysioNet/Computing in Cardiology Challenge 2019, training sets A and B (Reyna et al., *Crit Care Med* 2020)	**Coverage**	All 40,336 ICU stays, 1,552,210 patient-hours (no sampling)

Executive summary

Question. An early warning is only useful if staff still trust it. How much early catch does a standard vital-sign alert rule buy, how many alarms does the ward pay for it, and can a better rule or a simpler sensor change that trade?

Answer in one line. The standard NEWS2 ≥ 5 rule catches 6 in 10 sepsis patients about 10 hours early, but only about 1 in 34 alarms comes before a real sepsis case. Requiring the score to stay high for 2 hours (at a threshold of 4) keeps the same catch rate with 28% fewer alarms, and heart rate + breathing rate alone get within 2 points of the full five-vital score at the same alarm load.

59.7%
of sepsis patients alerted in the 12 h before onset by NEWS2 ≥ 5 (median 10 h early)

2.9%
of NEWS2 ≥ 5 alarms come before a sepsis onset (1.26 alarms per patient-day)

–28%
alarms with "score ≥ 4 for 2 hours", at the same 59.6% catch rate

57.6%
catch rate from heart rate + breathing rate alone at the NEWS2 ≥ 5 alarm load

1 · Data and method

Acquire
40,336 per-patient files from PhysioNet's open S3 mirror

→

Merge
One file with a patient id; header checked on every file

→

Query
5 numbered SQL files: window functions (LAG, LOCF), CASE scoring, grid CROSS JOIN

→

Check
Assert row and patient counts, monotonic trade-off curves, one cohort per rule

Cohort

Group	Stays	Use
No sepsis	37,404	Every hour scored; all their alarms are false for this question
Sepsis during the stay	2,506	Hours before onset scored; the 12 h before onset is the "catch window"
Septic on arrival	426	Excluded: nothing left to warn about
Total	**40,336**	**1,539,289 patient-hours scored**

PhysioNet shifts the sepsis label 6 hours early, so onset = first labelled hour + 6. Onset follows the Sepsis-3 definition.

Score and metric definitions

Term	Definition
Modified NEWS2	NEWS2 points for respiratory rate, SpO2, systolic BP, heart rate, temperature (0–15). No consciousness or oxygen-support fields in the data
HR + RR score	Only the heart-rate and breathing-rate points (0–6): what a contactless under-mattress sensor reads
Alarm	Fires when the score reaches the threshold (optionally for 2 hours in a row); a standing alarm counts once
Catch rate	Sepsis patients with an alarm on in the 12 h before onset
Precision	Alarms fired in a catch window ÷ all alarms

Vitals are carried forward from the last reading (what a monitor would still show). Never-measured vitals score 0.

2 · The alert-fatigue problem

The ward-standard trigger, **NEWS2 ≥ 5**, puts an alarm on **59.7%** of sepsis patients in the 12 hours before onset, with a median lead of **10 hours**. The cost is **1.26 alarms per patient-day**, and only **2.9%** of those alarms fire in a catch window. Put simply, a nurse answers about **34 alarms for every one** that comes before a sepsis case.

Rule (all 5 vitals)	Catch rate	Median lead	Alarms / patient-day	Precision
Score ≥ 3	88.6%	12 h	2.62	1.8%
Score ≥ 4	76.9%	11 h	1.94	2.3%
Score ≥ 5 (NEWS2 "medium")	**59.7%**	**10 h**	**1.26**	**2.9%**
Score ≥ 6	42.6%	9 h	0.73	3.4%
Score ≥ 7 (NEWS2 "high")	27.4%	8 h	0.37	4.2%

Median lead is capped at 12 h by the window. Below a threshold of 3 the alarm is on most of the time (38% of all hours at ≥ 3, 76% at ≥ 1), so it stops being a signal at all.

Raising the threshold buys precision slowly and loses catches quickly: going from ≥ 5 to ≥ 7 cuts alarms by 70% but loses more than half the catches, while precision only goes from 2.9% to 4.2%. Moving the threshold alone cannot fix alert fatigue.

3 · Smarter rules, same catch rate

A simple suppression rule waits until the score has been high for **2 hours in a row**. At the same threshold of 5, it cuts alarms by **58%** (1.26 → 0.53 per patient-day) but drops the catch rate by **18.8 points** (59.7% → 40.9%). The better use of persistence is to **lower the threshold and add the wait**:

Rule	Catch rate	Median lead	Alarms / patient-day
Score ≥ 5, fire at once (standard)	59.7%	10 h	1.26
Score ≥ 4, fire after 2 high hours	**59.6%**	**10 h**	**0.91 (–28%)**

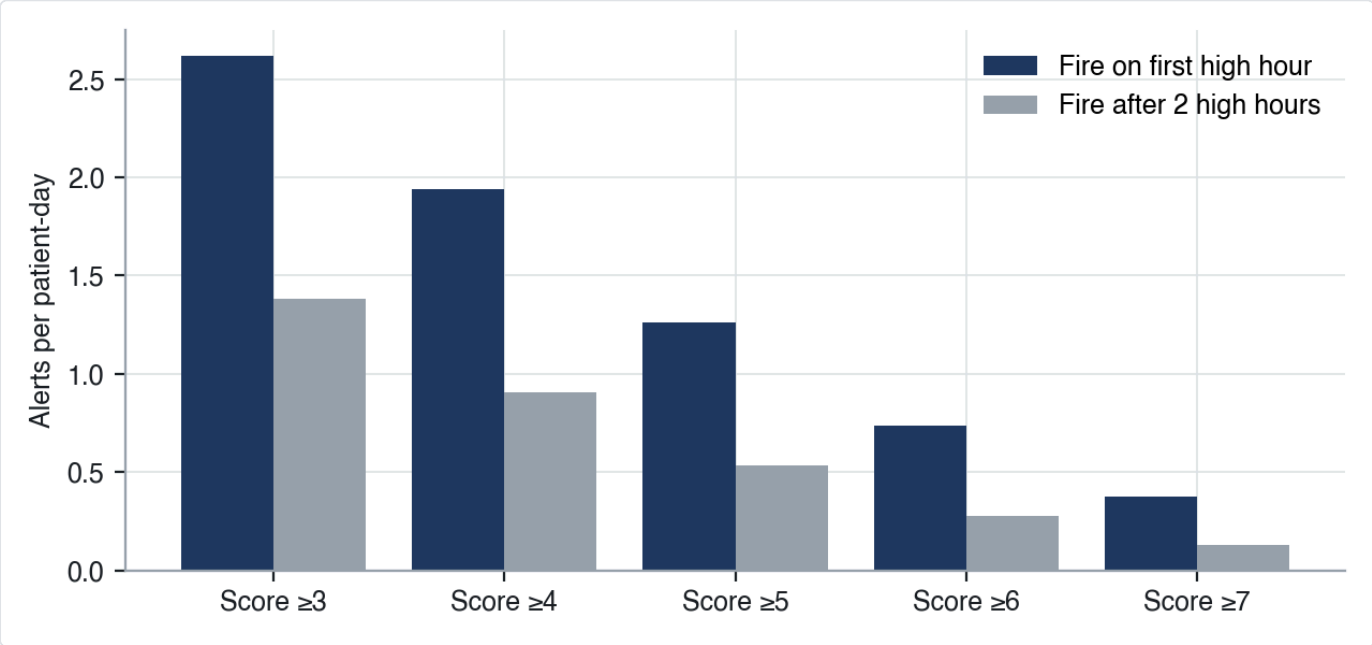


Figure 1 · Waiting for a second high hour cuts alarms by 47–66% depending on the threshold. Most single-hour spikes do not persist.

4 · Two vitals vs five

A contactless under-mattress sensor reads **heart rate and breathing rate**, continuously and without a nurse. Can those two alone stand in for the five charted vitals? Comparing the two scores **at the same alarm load** (the only fair comparison, because their scales differ): at NEWS2 ≥ 5's load of 1.26 alarms per patient-day, the HR + RR score catches **57.6%** vs **59.7%** for all five vitals. That is **96%** of the detection with two signals.

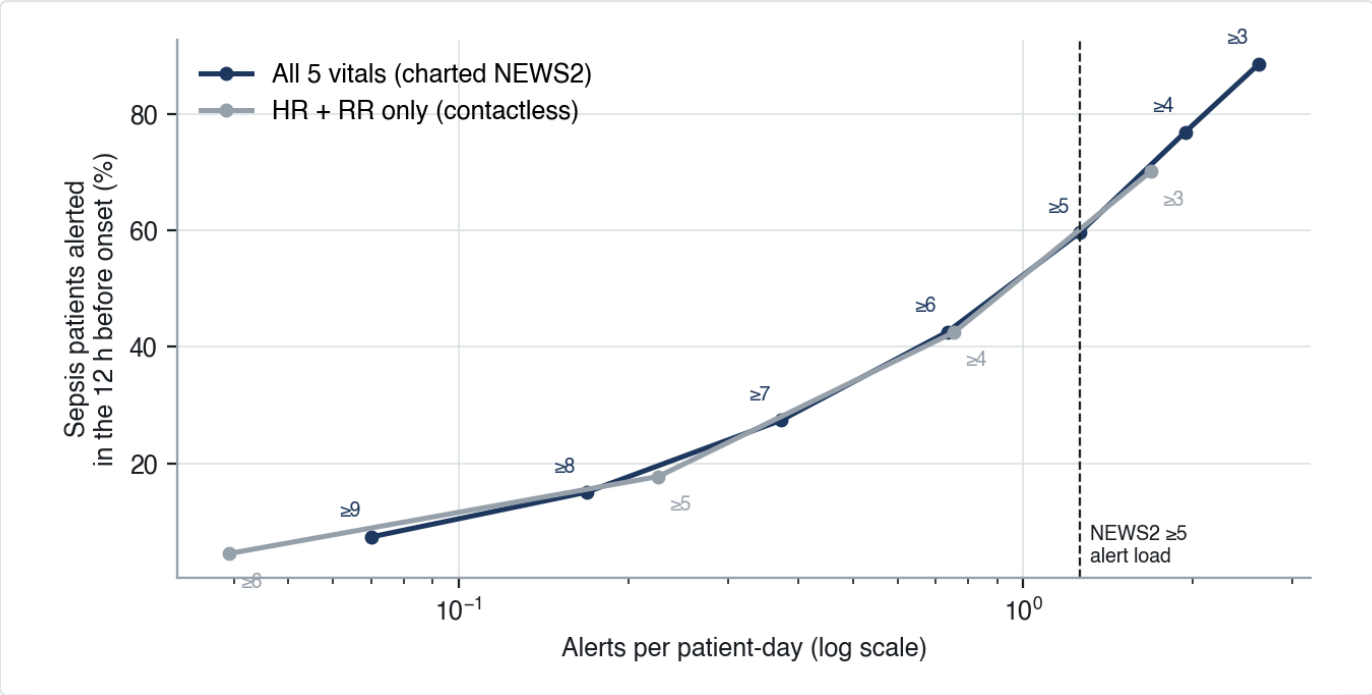


Figure 2 · The two trade-off curves sit almost on top of each other from 0.05 to 1.7 alarms per patient-day. Labels show each threshold.

Freshness matters too. Even in this ICU data, the "current" value is a fresh reading in only **90%** of hours for heart rate, **85%** for breathing rate and **34%** for temperature; the rest are carried forward. On a general ward, where vitals are charted every 4–6 hours, the gap is far larger. That is the case for continuous HR + RR, rather than for more vitals.

5 · What drives false alarms

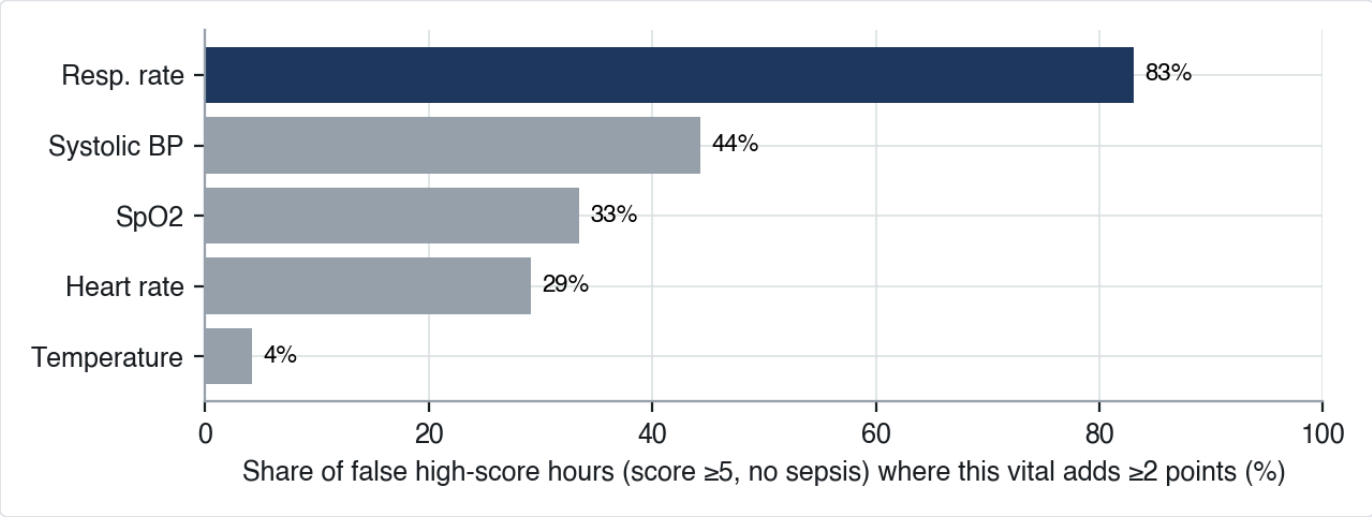


Figure 3 · In patients who never develop sepsis, breathing rate is involved in 83% of hours with a score of 5 or more.

Breathing rate is both the strongest signal and the biggest source of noise. For a product built on HR + RR, the accuracy and smoothing of the breathing-rate reading decide the alarm burden more than any threshold choice.

6 · Product recommendations (for a contactless monitoring product)

1. **Track one north-star metric with a guardrail:** deteriorations caught ≥ 6 h early per 100 bed-days (north star), against alarms per bed per day (guardrail). Report both in every hospital review, never one alone.
2. **Ship persistence plus a lower threshold, not a higher threshold.** In this data it keeps the catch rate and removes 28% of alarms. A/B it by ward: randomize the rule, and measure alarm acknowledgement time and missed escalations.
3. **Invest in breathing-rate quality first.** It drives 83% of false high scores. Gains there lower the alarm load for every rule.
4. **Sell continuity, not more vitals.** HR + RR keeps 96% of the five-vital catch rate at the same load, and is fresh every hour where charting is not.
5. **Instrument the nurse's side of the funnel:** alarm → seen → acknowledged → escalated → outcome. Precision of 3% is only tolerable if each alarm is cheap to clear.

7 · Limitations

- **ICU, not wards or homes.** ICU patients are sicker and charted hourly; ward and home rates will differ.
- **Sepsis stands in for "getting worse".** Some "false" alarms are real deterioration of another kind, so precision is a lower bound.
- **Retrospective scoring of charted values**, carried forward between readings. No real alarms or nurse responses were observed.
- **Modified NEWS2** without consciousness or oxygen support, so absolute scores run lower than the full NEWS2.
- **No Dozee data or claims are used.** "HR + RR" approximates what such a sensor reads; its real accuracy is not modelled.

Appendix · SQL files and reproducibility

File	What it does
`01_cohort.sql`	One row per stay; onset hour; no-sepsis / sepsis / septic-on-arrival
`02_hourly_scores.sql`	Carry-forward vitals (`last_value … IGNORE NULLS`), NEWS2 points, both scores, catch window
`03_rule_grid.sql`	30 rules (score × threshold × persistence); `LAG` for persistence and alarm on-switch; catch rate, lead, load, precision
`04_measurement_gaps.sql`	Share of hours with a fresh reading per vital
`05_false_alarm_drivers.sql`	Which vitals push no-sepsis patients to a score of 5+

```
# reproduce (~1 min after download)
./download_data.sh && python run.py   # results/*.csv, results/summary.json, charts/*.png
```

Data: Reyna MA et al. Early Prediction of Sepsis From Clinical Data: The PhysioNet/Computing in Cardiology Challenge 2019. *Crit Care Med* 2020;48(2):210–217. PhysioNet, CC BY 4.0. Analysis and any errors are the author's own.